

Worksheet 6.3

The Haber process

NAME:

CLASS:

INTRODUCTION

In 1908, Fritz Haber developed a process for synthesising ammonia from its elements. Until this time, Germany had been reliant on imported nitrates for the synthesis of fertilisers and explosives. With the outbreak of World War I, Germany's supply routes were blocked. Haber's process allowed Germany to remain in the war for much longer.

No.	Question	Answer
1	Write the thermochemical equation for the Haber process.	
2	How are the raw materials for the process obtained?	
3	Theoretically, what conditions would increase the <i>rate</i> of this process?	
4	Theoretically, what conditions would increase the <i>yield</i> of this process?	
5	What conflict exists between rate and yield considerations and how is this conflict resolved in practise?	
6	a Name the catalyst used in the Haber process. b Is this an example of a heterogeneous or a homogeneous catalyst?	

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No.	Question	Answer
7	How do catalysts speed up the rate of a chemical reaction?	
8	Draw a reaction profile for this reaction, showing the catalysed and uncatalysed pathways.	
9	a What is the maximum mass of ammonia that could be produced when 300 kg of nitrogen reacts with excess hydrogen? b What is the minimum mass of nitrogen needed to produce 5.68 tonnes of ammonia?	
10	Does the Haber process involve the oxidation or reduction of nitrogen? Use oxidation numbers to explain your answer.	
11	On a separate page, draw a flowchart to summarise the Haber process for ammonia production.	

Worksheet 6.5

The contact process

NAME:

CLASS:

INTRODUCTION

Sulfuric acid is the most widely produced synthetic chemical in the world, and is used extensively in the manufacture of fertilisers. Other uses include the removal of rust and scale, production of paints and synthetic dyes, manufacture of explosives, soaps and detergents, and use in polymer industries. Sulfuric acid is a strong acid, a strong oxidant and an excellent dehydrating agent. For these reasons, demand for it has always been high.

No.	Question	Answer
1	Sulfuric acid has been known for hundreds of years, and was originally called 'oil of vitriol' by the alchemists of the Middle Ages. Why would the alchemists have given it this name?	
2	Rather than using elemental sulfur as a key raw material for producing sulfuric acid, most plants in Australia now use sulfur dioxide gas. State two sources of SO_2 for this industry, showing relevant equations for its production (where appropriate).	
3	Write a thermochemical equation for the converter reaction occurring in the contact process.	
4	The sulfur dioxide and air must be thoroughly cleaned and dried before they are passed over the catalyst beds. Why is this preliminary step necessary?	
5	Explain the optimal conditions of temperature and pressure to produce the maximum theoretical yield of product in the contact process.	

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The contact process

No.	Question	Answer
6	In the contact process, a compromise must be made between rate and yield considerations. Describe the conditions actually used and why they are chosen.	
7	a Name the catalyst used in the contact process. b The catalyst is laid out in successive 'beds' across which the reactant gas mixture is rapidly passed. Explain why the catalyst is arranged in this manner in the context of both equilibrium and rate considerations.	
8	Explain why oleum ($\text{H}_2\text{S}_2\text{O}_7$) is generated as an intermediate step in the production of sulfuric acid, rather than the more direct hydrolysis of SO_3 with water.	
9	Using relevant equations to illustrate your answer, show why SO_2 and SO_3 are significant environmental pollutant gases.	
10	Draw a flowchart to summarise the contact process for sulfuric acid production.	

Worksheet 6.3: Solutions

The Haber process

No.	Answer
1	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) ; \Delta H = -92.3 \text{ kJ}$
2	Steam and methane are heated with air to high temperatures (between 700°C and 1100°C) and pressures (25–35 atm) in the presence of a nickel catalyst to produce carbon monoxide and hydrogen gas. The carbon monoxide formed is then reacted with more steam to produce carbon dioxide and more hydrogen gas: $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$ $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$ The carbon dioxide gas is absorbed by a suitable base, leaving a mixture of hydrogen and nitrogen (from the air) in the mole ratio 3:1.
3	Increased pressure, increased temperature, use of a catalyst and increased concentration of reactants would all increase its rate.
4	Increased pressure, decreased temperature, increased concentration of reactants and decreased concentration of product would increase its yield.
5	There is a temperature conflict. It is resolved by using a ‘compromise’ temperature of 450°C, which allows a satisfactory rate of reaction without compromising the yield too much. A catalyst is also used.
6	a Porous iron pellets with Al_2O_3 and KOH are used the catalyst. b It is a heterogeneous catalyst – the catalyst is solid but the reactants are gaseous.
7	They lower the activation energy by providing an alternative reaction pathway.
8	A similar profile to the one shown in Figure 6.6 on page 159 of the textbook.
9	a $n(\text{N}_2) = \frac{m}{M} = \frac{300\,000}{28.02} = 1.071 \times 10^4 \text{ mol}$ $n(\text{NH}_3) = 2 \times n(\text{N}_2) = 1.071 \times 10^4 \times 2 = 2.142 \times 10^4 \text{ mol}$ $m(\text{NH}_3) = n \times M = 2.142 \times 10^4 \times 17.034 = 3.65 \times 10^5 \text{ g}$ b $n(\text{NH}_3) = \frac{m}{M} = \frac{5.68 \times 10^6}{17.034} = 3.335 \times 10^5 \text{ mol}$ $n(\text{N}_2) = \frac{1}{2} \times n(\text{NH}_3) = \frac{1}{2} \times 3.335 \times 10^5 = 1.668 \times 10^5 \text{ mol}$ $m(\text{N}_2) = 1.668 \times 10^5 \times 28.02 = 4.67 \times 10^6 \text{ g}$
10	Oxidation number of N in N_2 is 0, while in NH_3 it is –3. A decrease in oxidation number is reduction.
11	See Figure 6.26 on page 188 of the textbook.

Worksheet 6.5: Solutions

The contact process

No.	Answer
1	It was named for its corrosiveness and extremely sour, or 'vitriolic', taste. In these times it was common for the alchemists to taste the chemicals that they had discovered or produced!
2	SO ₂ is produced as a: - by-product of the desulfurisation of petroleum in the petrochemical industries - waste product generated in the smelting of the sulfide ores of non-ferrous metals such as copper, zinc and lead. For example: $2\text{PbS(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{PbO(s)} + 2\text{SO}_2\text{(g)}$
3	$2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{SO}_3\text{(g)}; \Delta H = -197 \text{ kJ}$
4	It is necessary to prevent 'poisoning' of the catalyst. (This term indicates that the catalyst is rendered ineffective by dust or contaminants coating its surface.)
5	According to Le Châtelier's principle an increased yield can be achieved by: - increasing the gas pressure (3 moles of reactant gases produce only 2 moles of product gases, so an increase in pressure would enhance the forward reaction) - decreasing the temperature, as the forward reaction is exothermic.
6	Low temperatures ensure a significant yield of sulfur trioxide is formed, but slowly. At high temperatures, the equilibrium yield is poor, but it forms quickly. A compromise in temperature must be reached to achieve acceptable yields quickly. A temperature of about 440°C for the incoming gases is appropriate, and atmospheric pressure is used to save money on the costs associated with high pressure equipment.
7	a Vanadium(V) oxide or vanadium pentoxide b By using trays of pellets, the surface area of the catalyst is increased, and so is its effectiveness. Length of time for contact is carefully controlled because the temperature increases due to the exothermic reaction. At higher temperatures the SO₃ would quickly decompose to SO₂ and O₂.
8	$\text{SO}_3\text{(g)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{SO}_4\text{(aq)}$ While this direct hydrolysis of SO₃ would be more straightforward, it is highly exothermic and would result in the production of a 'fog' of sulfuric acid droplets. Oleum (H₂S₂O₇) is produced by reacting preformed sulfuric acid with SO₃. The oleum is then reacted with water: $\text{SO}_3\text{(g)} + \text{H}_2\text{SO}_4\text{(l)} \rightarrow \text{H}_2\text{S}_2\text{O}_7\text{(l)}$ $\text{H}_2\text{S}_2\text{O}_7\text{(l)} + \text{H}_2\text{O(l)} \rightarrow 2\text{H}_2\text{SO}_4\text{(aq)}$

Worksheet 6.5: Solutions

The contact process

No.	Answer
9	Both of these gases readily hydrolyse in rainwater to produce sulfurous and sulfuric acids respectively, which are major contributors to acid rain. Relevant equations are: $\text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_3(\text{aq})$ $\text{SO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_4(\text{aq})$
10	See Figure 6.30 on page 191 of the textbook.